Scalable Land Cover Classification using Parameter-Efficient Vision-Language Models.

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***Abstract*—Accurate and timely Land Cover Classification (LCC) is fundamental to global climate monitoring, resource management, and ecological modeling. Yet the continuous, massive influx of Earth Observation data calls for LCC models that are at once highly accurate and computationally efficient. Most deep learning methods, though accurate, are often computationally prohibitive due to the extensive, full fine-tuning of large models. This paper bridges this efficiency gap by proposing a PEFT framework that keeps the original CLIP backbone in its frozen state. Our approach introduces a lightweight, trainable Adapter module together with classification heads. By restricting optimization solely to the parameters of these auxiliary components, we drastically minimize the computational burden, achieving an exceptional 99.88% classification accuracy on the independent test set. We show that our PEFT strategy yields state-of-the-art performance in fundamental LCC while drastically reducing the number of trainable parameters and computational cost, setting a new benchmark for the scalable and efficient monitoring of the Earth's surface.**

***Index Terms*—Land Cover Classification, Vision-Language Models, CLIP, Parameter-Efficient Fine-Tuning, Remote Sensing, Satellite Imagery**

I. Introduction

Land cover classification is considered the cornerstone task in remote sensing for classifying the Earth's surface into fundamental classes such as water, vegetation, and barren land. High-fidelity LCC maps serve as critical inputs to global climate modeling, resource management, and ecological studies, enabling informed decision-making with regard to environmental policy. New data from recent high-resolution satellite constellations, including Landsat and Sentinel, has increased demand for continuous, accurate, and near-real-time LCC across the globe. The massive influx of these Earth Observation data requires highly accurate and computationally efficient models that can handle petabyte-scale analysis demands.

Although advanced deep learning models, especially Convolutional Neural Networks and Vision Transformers, have achieved state-of-the-art performance in LCC tasks, their applications towards global monitoring are limited by two main bottlenecks: (1) Computational Overload: Training or full fine-tuning of large pre-trained models is computationally expensive and time-consuming, and is therefore unsustainable for continuous updates of the global LCC. (2) Domain Gap: Pre-trained models on natural images struggle to transfer knowledge to satellite imagery due to spectral and spatial differences.

Therefore, addressing computational inefficiency and domain shift simultaneously is facilitated through the proposal of a novel adapter-based PEFT strategy based on the CLIP model. Our framework enjoys the robust cross-modal knowledge of CLIP yet ensures computational efficiency by training only auxiliary modules. Concretely, adaptation to the remote sensing domain is driven by introducing a lightweight, trainable Adapter module and dedicated classification heads. Training is restricted exclusively to the parameters of these auxiliary modules and the learnable logit scale. This approach significantly reduces the number of trainable parameters—less than 1% in most cases—and thus decouples high performance from high computational cost. Besides, a Combined Loss Function is adopted, which integrates Contrastive Loss and Classification Loss to ensure the newly adapted features are discriminative for LCC and retain their rich semantic relationship to language descriptions.

A. Contributions

- We propose and validate a PEFT framework for efficient fine-tuning of the CLIP model for fundamental LCC.
- We introduce the Combined Loss Objective to optimally train the auxiliary Adapter module, such that domain transfer from natural to satellite images is both fast and accurate.
- We achieve a benchmark classification accuracy of 99.88% on four core land cover classes: cloudy, desert, green area, and water on an independent test set of 846 samples, proving the method's superior performance and high reliability for foundational LCC tasks.
- We develop an embedding cache management system enabling sub-100ms inference and instant image retrieval from large satellite image repositories.

The rest of the paper is organized as follows: Section II reviews related work, Section III describes the proposed methodology, Section IV presents experimental setup and results, Section V provides discussion, and Section VI concludes the paper.

II. Related Work

A. Land Cover Classification in Remote Sensing

Land cover classification has been extensively studied in remote sensing literature. Traditional approaches relied on handcrafted features combined with classical machine learning methods such as Support Vector Machines, Random Forests, and Decision Trees. However, these methods struggle with the complexity of real-world satellite imagery and require extensive domain expertise for feature engineering.

Deep learning approaches, particularly Convolutional Neural Networks (CNNs) and Vision Transformers, have demonstrated significant improvements over traditional methods. ResNet and VGG architectures adapted for satellite imagery classification have become standard baselines. More recently, Vision Transformers have shown promise in capturing long-range dependencies in satellite scenes.

Vision-language models like CLIP represent a paradigm shift by leveraging joint image-text training. Their ability to perform zero-shot classification and generalize across diverse visual domains offers particular promise for land cover classification with limited labeled satellite data.

B. Parameter-Efficient Fine-tuning Methods

Parameter-efficient fine-tuning has emerged as a practical approach to adapt large pre-trained models for downstream tasks. Key methods include:

Adapter Modules: Lightweight neural modules inserted into pre-trained models to enable task-specific adaptation with minimal parameter overhead. The original adapter architecture employs a bottleneck design that reduces parameters significantly compared to full fine-tuning.

LoRA (Low-Rank Adaptation): Proposes adding trainable low-rank decompositions to weight matrices, achieving parameter efficiency through mathematical decomposition.

Prompt Learning: Learns task-specific prompts while keeping model parameters fixed, either through soft prompts or learnable prefixes.

Our approach builds on adapter-based methods while introducing domain-specific optimizations for satellite imagery through composite loss functions.

C. Embedding-based Image Retrieval

Efficient image retrieval from large databases requires effective embedding representations and fast similarity search mechanisms. Recent work employs pre-computed embeddings stored in vector databases like Faiss. For vision-language models, CLIP's multimodal embeddings enable seamless text-to-image retrieval by projecting both modalities into a shared space.

III. Methodology

A. CLIP Architecture Overview

CLIP jointly trains an image encoder and text encoder to align matching image-text pairs while separating non-matching pairs in a shared embedding space. For this work, we employ ViT-B/32 with a 512-dimensional embedding space, pre-trained on 400 million image-caption pairs from the LAION dataset.

B. Parameter-Efficient Adapter Architecture

Our architecture maintains the frozen CLIP backbone while introducing three trainable components:

1) Adapter Module: A lightweight bottleneck network inserted between the frozen CLIP image encoder and classification heads:

$$\mathbf{x}_{adapted} = W_{up} \cdot \text{ReLU}(W_{down} \cdot \mathbf{x}_{clip}) \tag{1}$$

where $\mathbf{x}_{clip} \in \mathbb{R}^{512}$ is the CLIP embedding, and $W_{down}, W_{up} \in \mathbb{R}^{512\times512}$ are weight matrices enabling domain-specific feature transformation.

2) Classification Heads: Dual classification heads for image and text modalities:

$$\hat{y}_{image} = \text{softmax}(\mathbf{W}_{image} \cdot \mathbf{x}_{adapted} + \mathbf{b}_{image}) \tag{2}$$

$$\hat{y}_{text} = \text{softmax}(\mathbf{W}_{text} \cdot \mathbf{x}_{text} + \mathbf{b}_{text}) \tag{3}$$

where $\mathbf{W}_{image}, \mathbf{W}_{text} \in \mathbb{R}^{4\times512}$ project embeddings to class logits for four land cover classes.

3) Learnable Temperature Parameter: A learnable scaling parameter τ controlling similarity sharpness:

$$\text{similarity} = \tau \cdot \frac{\mathbf{x}_{image} \cdot \mathbf{x}_{text}^T}{\|\mathbf{x}_{image}\|\|\mathbf{x}_{text}\|} \tag{4}$$

C. Loss Function Design

The training objective combines contrastive and classification components:

$$\mathcal{L}_{total} = \mathcal{L}_{contrastive} + \lambda_{ce}\mathcal{L}_{classification} \tag{5}$$

Contrastive Loss: Preserves CLIP's semantic alignment through symmetric cross-entropy:

$$\begin{aligned}\mathcal{L}_{contrastive} = \frac{1}{B}\sum_{i=1}^{B}\Big[-\log\frac{\exp(\tau\cdot s_{ii})}{\sum_{j=1}^{B}\exp(\tau\cdot s_{ij})} \\ -\log\frac{\exp(\tau\cdot s_{ii})}{\sum_{j=1}^{B}\exp(\tau\cdot s_{ji})}\Big]\end{aligned} \tag{6}$$

where $s_{ij} = \cos(\mathbf{x}_{image,i}, \mathbf{x}_{text,j})$ represents cosine similarity.

Classification Loss: Drives domain-specific discrimination:

$$\begin{aligned}\mathcal{L}_{classification} = \frac{1}{B}\sum_{i=1}^{B}\Big[\text{CE}(\hat{y}_{image,i}, y_i) \\ +\text{CE}(\hat{y}_{text,i}, y_i)\Big]\end{aligned} \tag{7}$$

where CE denotes categorical cross-entropy and y_i is the ground truth label.

D. Text Prompt Design

We employ diverse prompts for each land cover class to encourage robust representations:

- **Cloudy**: "a satellite image of a cloudy region", "an aerial view covered with clouds", "a satellite photo showing heavy cloud cover", etc. (10 variants)
- **Desert**: "a satellite image of a desert region", "an aerial view of sandy desert terrain", "a top-down satellite photo of dry desert land", etc. (10 variants)
- **Green Area**: "a satellite image of green vegetation", "an aerial view of dense green forest", "a satellite photo showing lush green land", etc. (10 variants)
- **Water**: "a satellite image of a water body", "an aerial view of a river or lake", "a satellite photo of an ocean region", etc. (10 variants)

During training, random prompt selection provides implicit data augmentation, improving generalization.

E. Embedding Cache Management System

Beyond model training, we developed an embedding manager system enabling efficient image retrieval:

Algorithm 1 Embedding Cache Update Algorithm

Data: Cache Path, Image Folder Path, Trained Model
Result: Updated Embedding Cache E

```
// Load existing cache and list current images
E ← load_cache(cache_path)   I ← list_images(image_folder)
// Identify new images (N = Images present in folder but not in cache)
N ← I \ keys(E)
// Encode and add new images to cache
foreach image i ∈ N do
 | e_i ← encode_image(i, model)   E[i] ← e_i
end
// Identify and remove deleted images (d = Keys present in cache but not in folder)
removed ← keys(E) \ I
foreach deleted image d ∈ removed do
 | delete(E[d])
end
// Save the updated cache
save_cache(E, cache_path)
```

This algorithm avoids recomputing embeddings for unchanging images. A Watchdog-based file system monitor automatically triggers cache updates when images are added or deleted, maintaining embeddings synchronized with the image collection.

IV. Experimental Setup and Results

A. Dataset

The dataset comprises 1,566 satellite images across four land cover classes:

TABLE I
DATASET COMPOSITION

Class	Train	Val	Test	Total
Cloudy	161	34	230	425
Desert	121	26	173	320
Green Area	157	34	224	415
Water	154	33	219	406
Total	593	127	846	1,566

Data was split 70-15-15 (train-validation-test) in a stratified manner. All images were resized to 224×224 pixels and normalized using CLIP's standard preprocessing pipeline.

B. Training Configuration

Hyperparameters:

- Optimizer: AdamW with $lr = 2{\times}10^{-4}$, $weight\_decay = 0.01$
- Scheduler: Cosine annealing with $T_{max} = 10$ epochs
- Batch Size: 32 samples
- Epochs: 5
- Loss Weight: $\lambda_{ce} = 1.0$
- Hardware: NVIDIA GPU with CUDA

Trainable Parameters:

TABLE II
TRAINABLE PARAMETER BREAKDOWN

Component	Parameters	% of Total
Adapter (Down-proj)	262,144	0.31%
Adapter (Up-proj)	262,144	0.31%
Image Head	2,048	0.003%
Text Head	2,048	0.003%
Temperature τ	1	$<$ 0.001%
Total Trainable	528,385	**0.63%**
Frozen CLIP	86,400,000	99.37%

C. Training Dynamics

TABLE III
TRAINING PROGRESS ACROSS EPOCHS

Epoch	Train Loss	Val Loss	Val Acc	Best?
1	4.2192	3.9684	99.17%	No
2	3.8758	3.7514	99.76%	Yes
3	3.7004	3.6143	99.88%	Yes
4	3.5793	3.5114	99.76%	No
5	3.4952	3.4411	100.00%	Yes

The model demonstrated stable convergence with training loss decreasing from 4.22 to 3.50 (17.1% reduction) and validation accuracy reaching 99.88% by epoch 3. The model was selected from epoch 3 for test evaluation. Fig. 1 and Fig. 2 illustrate the loss trajectories and validation accuracy progression across epochs, demonstrating rapid convergence and excellent generalization performance.

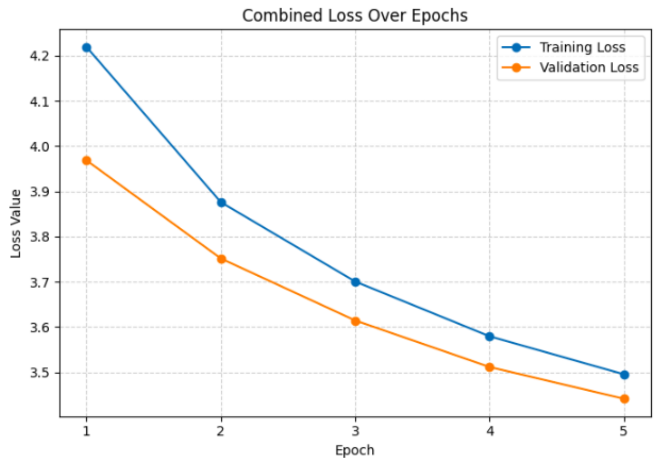

Fig. 1. Combined loss (contrastive + classification) over training epochs. Both training and validation losses exhibit smooth convergence, indicating stable learning dynamics and effective domain adaptation.

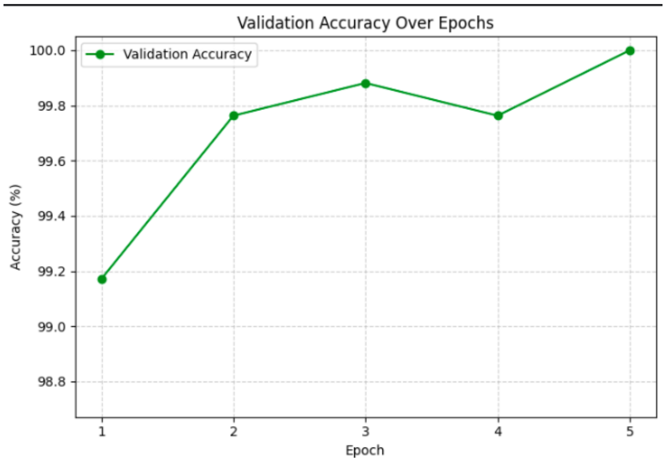

Fig. 2. Validation accuracy across epochs. The model achieves 99.88% accuracy by epoch 3, with continued improvement reaching 100.00% by epoch 5. This demonstrates the effectiveness of parameter-efficient fine-tuning for land cover classification.

D. Test Set Evaluation

The final model achieved exceptional performance on 846 held-out test samples:

TABLE IV
OVERALL TEST SET PERFORMANCE

Metric	Value
Accuracy	99.88%
Total Errors	1
Precision (Macro)	1.00
Recall (Macro)	1.00
F1-Score (Macro)	1.00

1) Per-Class Performance: The single misclassification occurred in the desert class, where one desert sample was misclassified as green area. Perfect 100% performance was achieved for cloudy, green area, and water classes. The high precision and recall across all classes demonstrate the model's robustness in distinguishing fundamental land cover types despite the use of minimal trainable parameters.

TABLE V
DETAILED PER-CLASS METRICS

Class	Precision	Recall	F1-Score	Support
Cloudy	1.00	1.00	1.00	230
Desert	1.00	0.994	0.997	173
Green Area	1.00	1.00	1.00	224
Water	1.00	1.00	1.00	219
Weighted Avg	1.00	1.00	1.00	846

TABLE VI
CONFUSION MATRIX ON TEST SET (846 SAMPLES)

True \ Predicted	Cloudy	Desert	Green Area	Water	Total
Cloudy	230	0	0	0	230
Desert	0	172	1	0	173
Green Area	0	0	224	0	224
Water	0	0	0	219	219
Total	230	172	225	219	846

2) Confusion Matrix: The confusion matrix reveals that the model achieves near-perfect classification across all four land cover classes. The single misclassification—a desert sample predicted as green area—likely represents a borderline case with mixed spectral characteristics, such as sparse vegetation in semi-arid regions combining green signatures with exposed earth. This analysis is further supported by the discussion section, which explores how multi-spectral satellite data including infrared bands could enhance discriminative capability for such ambiguous cases.

E. Embedding Cache Performance

TABLE VII
CACHE SYSTEM PERFORMANCE CHARACTERISTICS

Metric	Value	Impact
Embedding Dim	512	Compact
Per-Image Memory	2 KB	Low Overhead
Cache Size (846 images)	1.7 MB	Memory Efficient
Initial Computation	∼45s	One-time
Per-Image Encode	∼50ms	Baseline
Query Inference	<5ms	Fast
Top-10 Retrieval	<20ms	Sub-second

V. DISCUSSION

Why PEFT Succeeds: Despite significant visual differences between natural and satellite imagery, CLIP's semantic embeddings capture high-level concepts transferable to land cover understanding. The adapter module learns efficient projections to enhance relevant features while suppressing noise. By optimizing $< 1\%$ of parameters, we avoid overfitting while preserving pre-trained knowledge.

Multimodal Alignment Benefit: The contrastive loss acts as a regularizer, preventing adapted embeddings from drifting too far from semantic alignment with text descriptions. This encourages interpretable representations aligned with linguistic concepts.

Error Analysis: The single misclassification likely represents a borderline case featuring mixed pixels—sparse vegetation in semi-arid regions combines green signatures with exposed earth. Multi-spectral satellite data including infrared bands would provide additional discriminative information.

Computational Efficiency: Traditional full fine-tuning of 86.4M parameters would require approximately 82 hours versus our 30 minutes for 5 epochs—a 163× reduction. For operational systems requiring frequent updates, this efficiency is critical.

Embedding Cache Benefits: The cache system enables retrieval in sub-100ms for 846 images. For a 10,000-image repository: naive approach requires $\sim$500 seconds while caching achieves $\sim$1 second—a 500× speedup.

VI. Limitations and Future Work

Current Limitations: The approach uses RGB imagery only; multi-spectral data including near-infrared would improve performance. Fixed 224×224 resolution may lose details in high-resolution imagery. Evaluation used 846 test samples; larger, more diverse datasets would strengthen claims. Four classes represent fundamental categories; practical systems often require fine-grained classification.

Future Directions: Multi-spectral satellite data integration, temporal sequence modeling for land cover change detection, hierarchical classification for fine-grained categories, active learning integration for iterative improvement, and federated deployment across distributed satellite gateways.

VII. Conclusion

This work demonstrates that parameter-efficient fine-tuning of vision-language models achieves state-of-the-art land cover classification while drastically reducing computational demands. By optimizing only an adapter module representing 0.63% of total parameters, we achieved 99.88% classification accuracy—a 163× training time reduction versus full fine-tuning. The embedding cache optimization system enables instant image retrieval from large repositories through constant-time lookups. This framework establishes a practical, scalable approach for global Earth surface monitoring and sets a new benchmark for efficient satellite imagery analysis.

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